

Assessing Learned Models for Phase-only Hologram Compression

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Abstract

example abstract, **SIREN** [Sitzmann et al. 2020], and **FilmSIREN** [Chan et al. 2021], with **TAESD** [Min and Corso 2019]

CCS Concepts

• **Computing methodologies** → **Image compression**; • **Theory of computation** → **Data compression**; • **Hardware** → *Displays and imagers; Emerging optical and photonic technologies*.

Keywords

example

ACM Reference Format:

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1 Introduction

example introduction

2 Method

2.1 Optical Simulation via Physically-Based Rendering

The precise prediction of accommodation and vergence heavily depends on the quality of the synthesized ocular reflections. Historically, researchers have employed Zemax for such optical simulations; however, we found that traditional ray-tracing software is computationally exhaustive and yields suboptimal visual fidelity for dynamic Purkinje image generation.

To optimize both rendering time and optical effect, we constructed our simulation pipeline using Blender’s advanced rendering engine. By transitioning from conventional PCCR—which infers accommodation from fixation bias—to a DPI-centric approach within this modern rendering framework, we achieved superior optical sampling rates. The updated software allows for more efficient multi-path light tracing, enabling the extraction of high-resolution Purkinje images that accurately represent the eye’s physical accommodation state.

2.2 In-vivo Data Acquisition and Multimodal Inference

Building upon our established hardware framework, we extend our methodology to capture empirical in-vivo data from human subjects. To bridge this physical hardware with state-of-the-art computational paradigms, we conceptualize an intelligent diagnostic agent capable of automated clinical evaluation and the generation of plausible synthetic datasets for both medical and consumer applications. As a proof-of-concept for this integration, we formulate a novel inferential pipeline: raw optical captures acquired directly from the hardware are fed into a Multimodal Large Language Model (MLLM). Rather than relying solely on traditional analytical algorithms, we leverage the MLLM’s emergent reasoning capabilities to infer critical ocular properties, estimating physiological metrics such as vergence and dynamic accommodation states directly from the visual inputs.

3 Results and Discussions

example result

example table

References

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